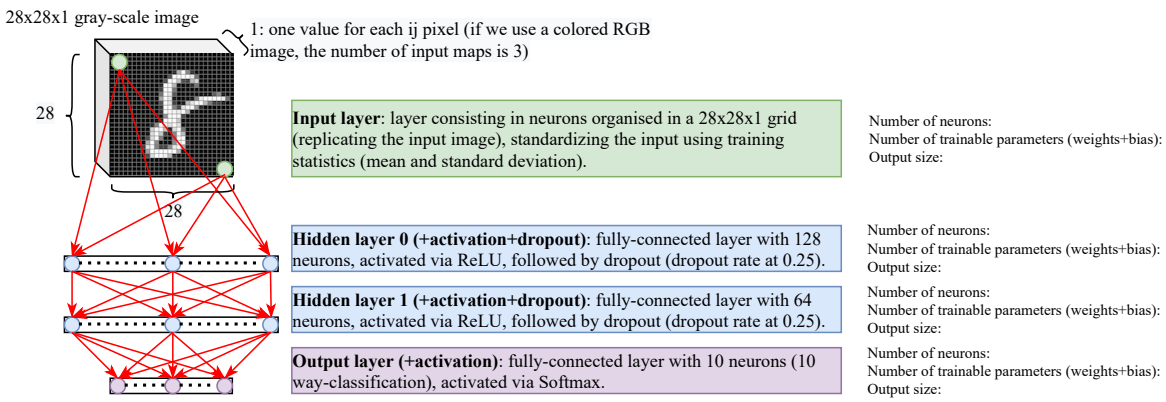
Neural Systems

(Academic year 2024-2025)

Exercises on supervised learning: Training a network with two hidden layers

Exercise 11

Handwritten digit classification is a common benchmark task addressed when designing deep neural networks with gray-scale images as input. The task consists in associating to a handwritten digit the correct class, i.e., $L = \{"0", \dots, "9"\}$ (10-way classification task). The most used dataset is MNIST (http://yann.lecun.com/exdb/mnist/), composed by 60000 training and 10000 test images 28x28.



- Data pre-processing and split. Load the MNIST dataset (both images and labels) and split it into training and test sets. Scale images from 0-255 to 0-1 range. Add a singleton dimension (28x28→28x28x1) as the input feature maps have only 1 input channel, corresponding to the grey-scale image. Apply one-hot encoding to labels by designing a proper function (e.g., named *to_one_hot*).
- Network design. Study the network reported in the figure. Compute the number of neurons, trainable parameters, and the output size for each layer (report them in the figure above). Then, define the network, by stacking the layers with the desired order. Inspect the defined network and check the number of trainable parameters / output size (previously found).
- Network training. Extract a validation set from the overall training set (20% of the training set). Train the network by minimizing the cross-entropy loss function using stochastic gradient descent with momentum, using: a learning rate=0.001, momentum term=0.9, batch size=128, max epochs=100. Track also the accuracy in addition to the loss during training, on both training and validation sets. Save only the best model on the validation loss, overall, during training (early stopped model). Plot the training and validation losses (/accuracies) over epochs.
- Network evaluation. Load the saved best model (corresponding to the early stopped model). Evaluate the model (i.e., compute the predicted output probabilities) on the test set. Compute the predicted labels as the labels with the highest output probability. Visualize a random test image and inspect the corresponding network output probabilities, predicted label and true label. Compute the confusion matrices and accuracies on both training, validation, and test sets.

Notes. The functions and classes needed to solve the exercise are described in the lecture note "An introduction to deep learning toolkits for supervised learning".